Downloaded from gsabulletin.gsapubs.org on October 28, 2010

Geological Society of America Bulletin

Paleomagnetism and paleogeography of Jurassic radiolarian cherts from the northern Apennines of Italy

I.W. Aiello and J.T. Hagstrum

Geological Society of America Bulletin 2001;113;469-481
doi: 10.1130/0016-7606(2001)113<0469:PAPOJR>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

Paleomagnetism and paleogeography of Jurassic radiolarian cherts from the northern Apennines of Italy

I.W. Aiello*
Consiglio Nazionale delle Ricerche, Centro di Studio dell'Appennino e delle Catene Perimediterranee, Florence 50121, Italy

J.T. Hagstrum†
U.S. Geological Survey, Menlo Park, California 94025, USA

ABSTRACT

Oriented samples of Jurassic radiolarian chert were collected from the Tuscan domain (continental margin) and the Ligurid domain (oceanic) of the northern Apennines for paleomagnetic study to determine the paleogeographic origins of these rocks. The oceanic rocks are all thermochemically overprinted by a mostly reversed-polarity component of magnetization (B) that was likely acquired during late Miocene regional uplift of the northern Apennines. This component also dominates the lower brittle chert of the Tuscan Cherts, but disappears upsection in the more clay-rich and ductile siliceous marlstones. In addition, the Tuscan Cherts retain an inferred primary magnetization (C), isolated at temperatures between 560 and 660 °C, which passes a fold test and shows a polarity stratigraphy. This component indicates a paleolatitude of 11° ± 4°N, and a counterclockwise vertical-axis rotation of 29° ± 9° with respect to the southern Alps of Italy, of 49° ± 8° with respect to Africa, and of 91° ± 8° with respect to Eurasia. Our results suggest that the Tuscan domain was farther south than other deep-water continental margin sections of Adria, and that transcurrent faulting might have played a significant role in the orogenic evolution of the northern Apennines.

Keywords: Jurassic, northern Apennines, paleogeography, paleomagnetism, radiolarite, western Tethys.

INTRODUCTION

In this study we present the first results of our paleomagnetic and stratigraphic investigation of Mesozoic radiolarian cherts in the Mediterranean region (Fig. 1). This paper focuses on the paleomagnetism of Middle to Upper Jurassic radiolarian cherts in the northern Apennines of Italy. The radiolarian cherts are usually found in continental margin sections of the Tuscan and Umbrian domains, and in the sedimentary cover of ophiolitic rocks from the Ligurian-Piedmontese ocean (Ligurid units; Fig. 2). Sections of these two facies were juxtaposed during convergence and partial accretion of the western Tethys oceanic plate to the Corsica-Sardinia plate between Eocene and late Miocene time. In particular, the subjects of this investigation are the Tuscan Cherts, within the continental margin of the Tuscan domain, and the Monte Alpe Cherts, which overlie ophiolites of the Ligurian domain. Although remagnetization has obliterated the primary remanent magnetization within the ophiolitic sections, the combined utilization of paleomagnetism and detailed lithostratigraphy has provided new information on the paleogeography and tectonic history of the Tuscan Cherts, as well as on vertical-axis rotation of the Tuscan block with respect to other continental margin domains of western Tethys.

PREVIOUS WORK

Deposition and Diagenesis of Cherts

The bedding of the northern Apennines cherts is typically rhythmic and characterized by regular alternations between chert beds and claystone interbeds. The chert-shale couplet might record primary variations either in radiolarian plankton productivity, or in pulses of clay input from continental sources. Many of the chert-shale couplets, however, are products of turbidity currents and reworking of sediment by deep-water currents, suggesting a rugged seafloor morphology during deposition. The primary compositional and textural properties of the sediment were subsequently enhanced by diagenesis (Nishbet and Price, 1974; Folk and McBride, 1978; Kälin et al., 1979; McBride and Folk, 1979; Barrett, 1982; Decker, 1990). Studies of diatomaceous deposits of the Monterey Formation in California (e.g., Pisciotto, 1981) have demonstrated that lithification of biosiliceous sediments is a time-temperature−dependent process in which amorphous silica of the tests (opal-A) undergoes recrystallization to quartz with opal-CT as an intermediate phase. The opal-A to opal-CT phase change occurs earlier in sediments composed of almost pure biogenic silica, whereas transformation of opal-CT to quartz is apparently retarded in sediments proportional to the clay content (Kastner et al., 1977; Kastner and Gieskes, 1983).

Examples of early diagenesis producing semilithified protobeds of biosiliceous sediments are common in the modern oceans (Bohrmann et al., 1994), in the Cenozoic diatomaceous formations of California, and in the Jurassic cherts of western Tethys (Jenkyns and Winterer, 1982). As we discuss herein, early diagenesis of silica and precipitation of hematite are two fundamental characteristics of radiolarian cherts that make them well suited for paleomagnetic investigations. Moreover, by using a new radiolarian biozonation for the Mesozoic (Baumgartner et al., 1995), it has been possible to obtain more accurate age determinations for chert sections within the northern Apennines (Chiari et al., 1997).

In the modern oceans there appears to be abundant growth and a rich diversity of bio-

*Present address: Department of Ocean Sciences, University of California, Santa Cruz, California 95064, USA; e-mail: iwaiello@caucsc.edu.
†Corresponding author; e-mail: jhag@usgs.gov.

GSA Bulletin; April 2001; v. 113; no. 4; p. 469–481; 9 figures; 3 tables.

For permission to copy, contact Copyright Clearance Center at www.copyright.com
© 2001 Geological Society of America

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

siliceous plankton (diatoms, radiolaria) at very high and at low latitudes. In particular, radiolarians are concentrated in the warm waters of the tropical and subtropical equatorial zones (e.g., equatorial Pacific and Atlantic Oceans, Caribbean Sea, and western Indian Ocean), although radiolarians also proliferate in northern water masses (Anderson, 1983). Due to the high degree of silica undersaturation in the oceans, the preservation potential for radiolarians increases in sedimentary basins that are located beneath zones of high biologic productivity and beneath strong current systems (e.g., equatorial currents and Gulf Stream; Goll and Bjørklund, 1971; Heath, 1974). Deep currents, however, can displace microfossils over considerable distances (Anderson, 1983).

The distribution of modern biosiliceous sediments indicates that high biologic productivity, associated with upwelling of nutrient-rich waters, was a likely requirement for deposition and preservation of radiolarian-rich sediments in western Tethys (Garrison, 1974; Baumgartner, 1987). Deposition of these sediments occurred in relatively deep basins below the local calcite compensation depth (CCD; Garrison and Fischer, 1969; Bosellini and Winterer, 1975; Barrett, 1982; Jenkyns and Winterer, 1982). However, Bosellini and Winterer (1975; see also Winterer, 1998) estimated that the CCD during Bajocian to Bathonian time in the western Tethys ocean was ~2500 m, which is shallower than the CCD of the modern oceans (which ranges between 3000 and 5000 m). As pointed out by Baumgartner (1987), the observed small-scale facies patterns of coeval cherts and pelagic limestones may be better explained by variations in the rates of general biologic productivity and by local redeposition of calcareous and siliceous sediments, than solely by a shallower CCD in Jurassic time. Moreover, deposition of Tethyan radiolarites might have occurred in a much greater depositional range than in deeper waters alone, as indicated by some localities where radiolarian chert beds are associated with calcareous beds of shallow-water origin (Molina et al., 1999).

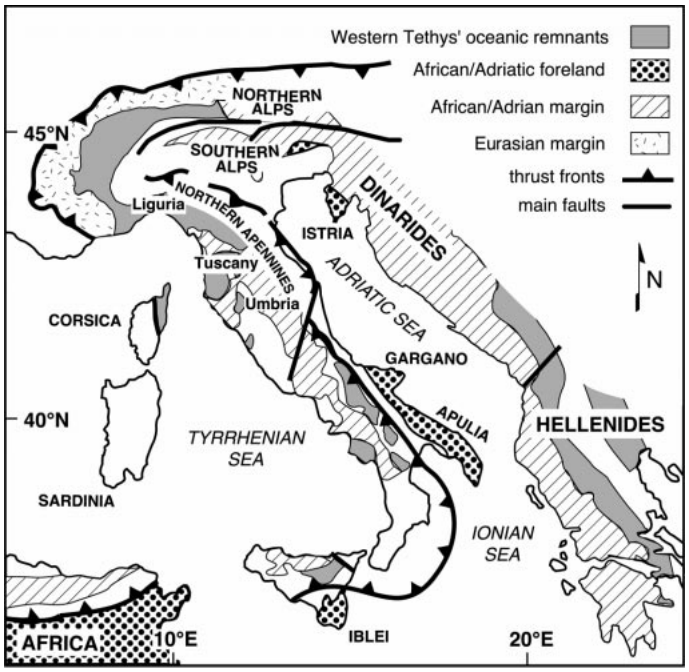


Figure 1. Geologic and tectonic setting of the western Mediterranean region (modified from Channell, 1996).

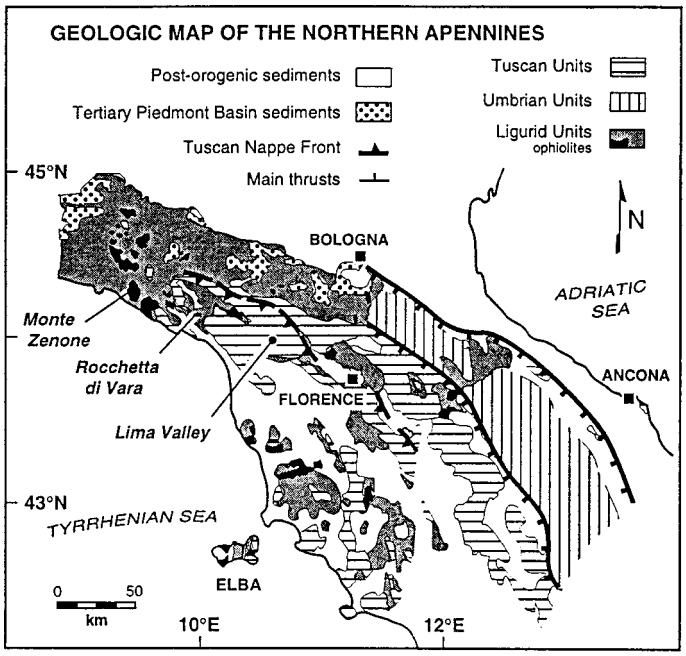


Figure 2. Distribution of tectonic units in the northern Apennines of Italy, and location of the studied chert sections (modified from Abbate et al., 1980).

Paleomagnetism

The first paleomagnetic investigation of red radiolarian chert (Middle Triassic to Upper Jurassic) that isolated a clear primary component of magnetization was that of Shibuya and Sasajima (1986) in the Mino terrane of central Japan. The high unblocking temperature component was interpreted as having been acquired near the time of deposition because it showed both positive and negative inclinations (normal and reversed polarities), passed both tectonic and intraformational fold tests, and indicated a depositional paleolatitude of 1° ± 3°N or S. Similar results were obtained for red Mesozoic cherts along the western margin of North America in California (Hagstrum and Murchey, 1993; Hagstrum et al., 1996) and western Mexico (Hagstrum and Sedlock, 1992). The paleomagnetic data and plate reconstruction models indicated that the western North American rocks were deposited

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

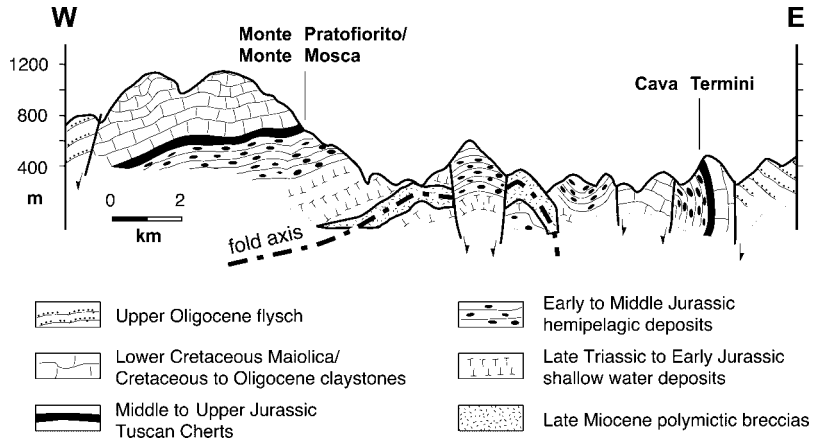


Figure 3. Geologic cross section of the Lima Valley (modified from Fazzuoli et al., 1998). The dashed line indicates the axis of the main recumbent fold with northeastern asymmetry. Folding is younger than late Miocene polymictic breccias within the fold.

together on the Farallon plate as the depositional site moved eastward along the paleoequator (0°–2°N or S paleolatitude) between Pliensbachian and Oxfordian time. Hagstrum (1990) determined that previous paleomagnetic studies in the Franciscan accretionary complex mostly dealt with completely remagnetized rocks, and subsequent work (Hagstrum et al., 1996; Hagstrum and Murchey, 1996) showed that among accreted oceanic rocks of the California margin, red (hematite bearing) radiolarian cherts most reliably retain a primary component of remanent magnetization.

No paleomagnetic investigations of Mesozoic rocks have been previously undertaken in the northern Apennines of Tuscany. Channell (1996) reviewed paleomagnetic data from other localities of the western Mediterranean region relevant to our study. Unfortunately, the Mesozoic African apparent polar wander path (APWP) is not well defined, and paleomagnetic poles from other continents have been rotated into African coordinates to provide a synthetic APWP for this time interval (Besse and Courtillot, 1991; Van der Voo, 1993). The Jurassic poles for North America are important to this synthesis, but remain inconsistent and controversial (Hagstrum, 1993; Van Fossen and Kent, 1990).

Jurassic and Cretaceous paleomagnetic data for the southern Alps of Italy (Fig. 1) were obtained for the Senonian, Hauterivian-Barremian, Kimmeridgian-Tithonian, and Callovian-Oxfordian by Channell et al. (1992a). Comparisons of the resultant APWP for the southern Alps and Umbria with the synthetic African APWP of Besse and Courtillot (1981) show that the Lower Cretaceous poles are generally similar to the African poles and that the Jurassic poles are offset, indicating vertical-axis rotations between the two regions. Declinations for coeval rocks in Umbria-Marche and the southern Alps also differ, indicating an ~20° counterclockwise rotation of Umbria with respect to the southern Alps (Channell, 1992), which is probably due to local tectonic rotations of thrust sheets (Channell, 1996). Cretaceous poles from Istria, Apulia, Gargano, and Iblei (Fig. 1) are also indistinguishable from the African APWP (Van der Voo, 1993; Channell, 1996). According to Channel et al. (1992b), the Lower Jurassic Adnet Limestone of the central northern Calcareous Alps carries a well-defined and apparently primary component of magnetization in hematite, and paleomagnetic poles from this unit are rotated between 30° and 100° clockwise with respect to coeval African poles.

GEOLOGIC AND TECTONIC SETTING

The opening of western Tethys resulted from sinistral east-west strike-slip motion between the African and Eurasian plates (Dercourt et al., 1986), and the oceanic spreading phase was controlled largely by transtensional mechanisms that formed pull-apart basins (Abbate et al., 1980, 1994). While there is widespread agreement concerning the tectonic setting of the opening of the western Tethys, the location and number of intervening microplates (collectively referred to as Adria) and oceanic corridors remain a matter of debate (for further discussion see Abbate et al., 1986; Robertson et al., 1996).

The radiolarian cherts (Monte Alpe Cherts) of the northern Apennine ophiolites (Ligurid units) represent the early sedimentary cover of the Ligurian-Piedmontese oceanic basin, which had a spreading axis (oriented approximately northeast-southwest) segmented by several transform fault zones (Abbate et al., 1980). Some of these faults might also have extended onto adjacent continental margins, possibly forming what is now the Serchio-Lima trough in northern Tuscany (Fazzuoli and Sguazzoni, 1986). The northern Apennine ophiolites are subdivided into a basement of serpentinized peridotites and subordinate gabbroic rocks (residual mantle) and an overlying cover of ophicalcites, ophiolitic breccias, basalts, radiolarian cherts (Monte Alpe Cherts), pelagic carbonates (Calpionella Limestone), and pelagic shales (Palombini Shales; Cortesogno et al., 1978; Abbate et al., 1980). While the basalt composition is mid-ocean ridge basalt (MORB) type, the extensive occurrence of mantle peridotites, the lack of a well-developed dike complex, the modest thickness of basalt flows, and an abundance of ophiolitic breccias indicate a different genetic model than that for standard oceanic crust (Abbate et al., 1994). Ascent of mantle material at a very slow spreading oceanic ridge (Abbate et al., 1994) and mantle exposure due to crustal delamination (Lemoine et al., 1987) are two of the possible mechanisms suggested for generation of oceanic crust in the Ligurian-Piedmontese ocean.

Whether the Ligurian-Piedmontese ocean had standard oceanic crust, the sedimentary facies deposited on western Tethys continental margins are typical of passive-margin settings, and are similar to the Mesozoic passive-margin facies of the central Atlantic (Bernoulli, 1972; Bernoulli and Jenkyns, 1974; Baumgartner, 1987). Truly marine sedimentation over the continental margins began in Triassic time, and included the buildup of reefs and carbonate platforms (Fig. 3). In Adria's tectonically stable regions (e.g., central Apennines, Istria, Gargano, Iblei, and Apulia; Fig. 1), shallow-water sedimentation continued throughout the Mesozoic. In those areas affected by tectonic wrenching, caused by movements between the African and Eurasian plates (e.g., southern Alps, Tuscan and Umbrian domains), breakup of the Triassic carbonate platforms was followed by deposition of pelagic sediments on subsiding continental margins. Because of the uneven seafloor topography, condensed pelagic sequences (e.g., Rosso Ammonitico facies) were deposited on topographic highs while thicker and resedimented strata (e.g., Calcare Selcifero della Val di Lima; Fig. 4) accumulated in topographic depressions that trended northeast-southwest in northern Tuscany.

A pronounced lull in tectonic activity between the middle Bathonian and Callovian, af-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

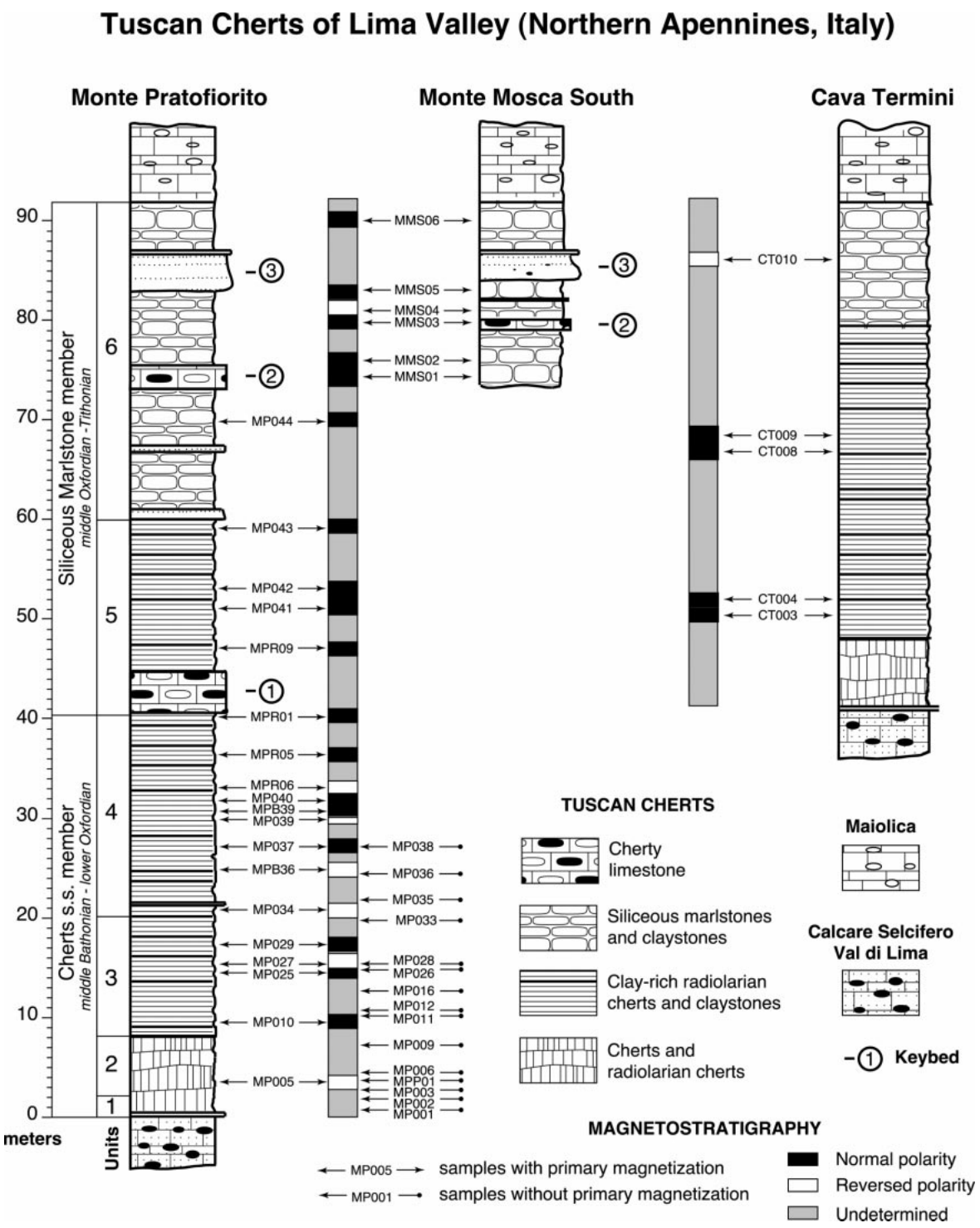


Figure 4. Stratigraphic columns of the Tuscan Cherts at the Monte Pratofiorito, Monte Mosca South, and Cava Termini sections (Lima Valley, northern Tuscany). In Lima Valley, the Tuscan Cherts overlie turbiditic cherty limestones of the Calcare Selcifero della Val di Lima and are overlain by the Maiolica Formation. The lower Chert sensu stricto member at Monte Pratofiorito is divided into 4 units: unit 1, an ~2-m-thick interval of vitreous, lenticular, and dark green chert; unit 2, red radiolarian chert beds interbedded with paper-thin claystones; and units 3 and 4, red to dark brown radiolarian chert beds, rhythmically alternating with claystones. The Siliceous Marlstone member is divided into two units (5 and 6). The most common lithology is red to green siliceous marlstone (unit 6). Upsection, the red siliceous marlstone beds become pinkish, with common green bands and laminae. Key beds 1 and 2 are cherty limestone horizons; key bed 3 is a massive siliceous marlstone bed.

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 1. PALEOMAGNETIC DATA OF REMANENCE COMPONENTS FOR TUSCAN AND LIGURIAN CHERTS

Section	I (°)	D (°)	N/n	R	k	α_{95} (°)	λ_{VGP} (°N)	ϕ_{VGP} (°E)
Component A								
Monte Pratofiorito								
Monte Mosca South								
Cava Termini	67.0	352.5	38/54	48.6694	37	3.4	82.4	330.6
Monte Mosca North	59.4	34.7	3/3	2.9654	58	16.4	64.2	96.7
Figline	58.7	357.4	8/8	7.8877	62	7.1	85.1	215.4
Rocchetta di Vara								
Monte Zenone	58.9	358.6	10/19	9.1420	10	15.7	85.3	203.5
Component B								
Monte Pratofiorito								
Monte Mosca South								
Cava Termini	−75.4	358.6	29/54	28.0825	8	9.7	16.6	11.0
Monte Mosca North	45.0	242.6	3/3	2.9908	218	8.4	0.9	317.7
Figline	62.2	280.6	8/8	7.8926	65	6.9	35.0	310.6
Rocchetta di Vara								
Monte Zenone	−45.8	166.0	14/19	13.1668	16	10.4	69.5	227.8
Component C								
Monte Pratofiorito								
Monte Mosca South								
Cava Termini	21.0	281.2	33/54	30.3206	12	7.6	16.0	280.9

Notes: I and D are inclination and declination, components A and B are in situ, and C is corrected for structural tilt; N/n is number of samples/specimens analyzed; R is vector sum of N unit vectors; k is the concentration parameter (Fisher, 1953); α_{95} is the radius of 95% confidence; λ_{VGP} and ϕ_{VGP} are latitude and east longitude of virtual geomagnetic pole (VGP). Sampling localities: Monte Pratofiorito (44.05°N, 10.28°E), Monte Mosca (44.08°N, 10.26°E), Cava Termini (44.04°N, 10.32°E), Figline (43.92°N, 11.10°E), Rocchetta di Vara (44.37°N, 9.76°E), and Monte Zenone (44.44°N, 9.51°E).

fecting both oceanic and continental margin areas, coincided with deposition of many of the radiolarian chert facies in the northern Apennines. Pelagic sedimentation of carbonate-free, radiolarian-rich sediments began within the Tuscan domain by the end of the Middle Jurassic (middle Bathonian; Chiari et al., 1997; Cortese, 1995). The Tuscan Cherts are divided into two informal members (Fig. 4): the lower Chert sensu strictu member and the upper Siliceous marlstone member (Aiello, 1994). The two members can be correlated, respectively, with the Ribbon Radiolarites of the southern Alps (Bosellini and Winterer, 1975), and the Rosso ad Aptychi identified in the southern Alps by Bosellini and Winterer (1975) and in southern Tuscany by Kälin et al. (1979). The maximum age of the Monte Alpe Cherts appears to be older in Liguria, where it is middle Bathonian to early Callovian at Rocchetta di Vara and Monte Zenone (Aiello and Chiari, 1997), than in Tuscany, where it is younger than late Oxfordian at Figline di Prato just northwest of Florence (Fig. 2; Chiari et al., 1997).

Beginning with the Oxfordian, radiolarian-rich sediments became gradually mixed with fine-grained terrigenous and very fine grained calcareous sediments, and by latest Jurassic time the predominant lithology was siliceous marlstone (Siliceous Marlstone member; Fig. 4). At the boundary between the Jurassic and Cretaceous, a major paleoceanographic change occurred and siliceous sedimentation was replaced by carbonate-rich pelagic sedimentation dominated by calcareous phytoplankton. These pelagic limestones, ubiquitous in western Tethys and the central Atlantic, are the main rock type of the Maiolica Formation (late Tithonian to Berriasian; Conti and Marcucci, 1986), which is coeval with the Ligurid domain's Calpionella Limestones. Both the Tuscan and Monte Alpe Cherts sections, however, show a steadily increasing amount of clay input during the Oxfordian to Tithonian (Fig. 4) that probably reflects a resurgence of tectonic activity in nearby landmasses and incipient orogenesis of the Dinaric-Hellenic chain. Thus, the lithostratigraphies of the Tuscan and Monte Alpe Cherts show major similarities that imply a common, large-scale, evolutionary trend for pelagic sedimentation in western Tethys during Middle to Late Jurassic time.

Accretion of ophiolitic (Ligurid) units within the northern Apennines occurred in Late Cretaceous to Eocene time (Dercourt et al., 1986). The compressional phase for the Dinarides-Hellenides (Late Jurassic; Robertson et al., 1996) and northern Calcareous Alps (late Middle Jurassic; Gawlick et al., 1999), however, occurred earlier. The northern Apennines chain has been regarded as an accretionary prism that formed during the orthogonal convergence between the European and the Adrian plates. During Miocene time, collision of the Adrian and Corsica-Sardinian plates caused eastward overthrusting of the Tuscan units by the Ligurid units, which together overthrusted the more external Umbrian units (Fig. 2; Boccaletti et al., 1971; Boccaletti et al., 1980; Principi and Treves, 1984). Alternatively, Marroni and Treves (1998) suggested that some large-scale features of the northern Apennines (e.g., lack of a magmatic arc) can be better explained in terms of strike-slip movements causing transpressional convergence.

The Lima Valley (Figs. 2 and 3), our main sampling locality of Tuscan Cherts, is located near the front of the Tuscan nappe. The dominant structural feature of this area is a large recumbent fold with northeastern asymmetry, which formed in two coaxial phases during the late Miocene (Fig. 3; Fazzuoli et al., 1998). Earlier extensional tectonics had caused numerous dip-slip faults, predominantly with northwest-southeast trends.

PALEOMAGNETIC METHODS AND RESULTS

Cherts are too durable to drill in the field, so oriented hand samples were collected from 42 beds at Monte Pratofiorito (Figs. 2 and 3), 6 beds at Monte Mosca South, 13 beds at Monte Mosca North, 6 beds at Cava Termini with a significantly different bedding attitude than that at Monte Pratofiorito, 8 beds at Figline di Prato, 12 beds at Rocchetta di Vara, and 7 beds at Monte Zenone. Red cherts were selectively sampled, and zones of alteration and recrystallization were avoided; bedding attitudes were taken for each bed collected. The hand samples were drilled in the laboratory with a water-cooled drill press and the core samples were cut into specimens 2.5 cm in length. Specimens were subjected to progressive thermal demagnetization in a magnetically shielded oven (internal field <2 nT) to temperatures <680 °C and their remanent magnetizations were measured using a cryogenic magnetometer, both instruments being within a shielded room-size enclosure. Lines representing the characteristic and overprint magnetization directions were fitted to the demagnetization end points using a least squares method based on principal component analysis (Kirschvink, 1980). The statistics of Fisher (1953) were used in analyzing the mean directions and poles given in Table 1.

Overall, two components of remanent magnetization were isolated in the specimens from the Monte Alpe Cherts and three components were isolated in the Tuscan Cherts. The first component (A) removed from almost all specimens of both formations (Table 1), usually by

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

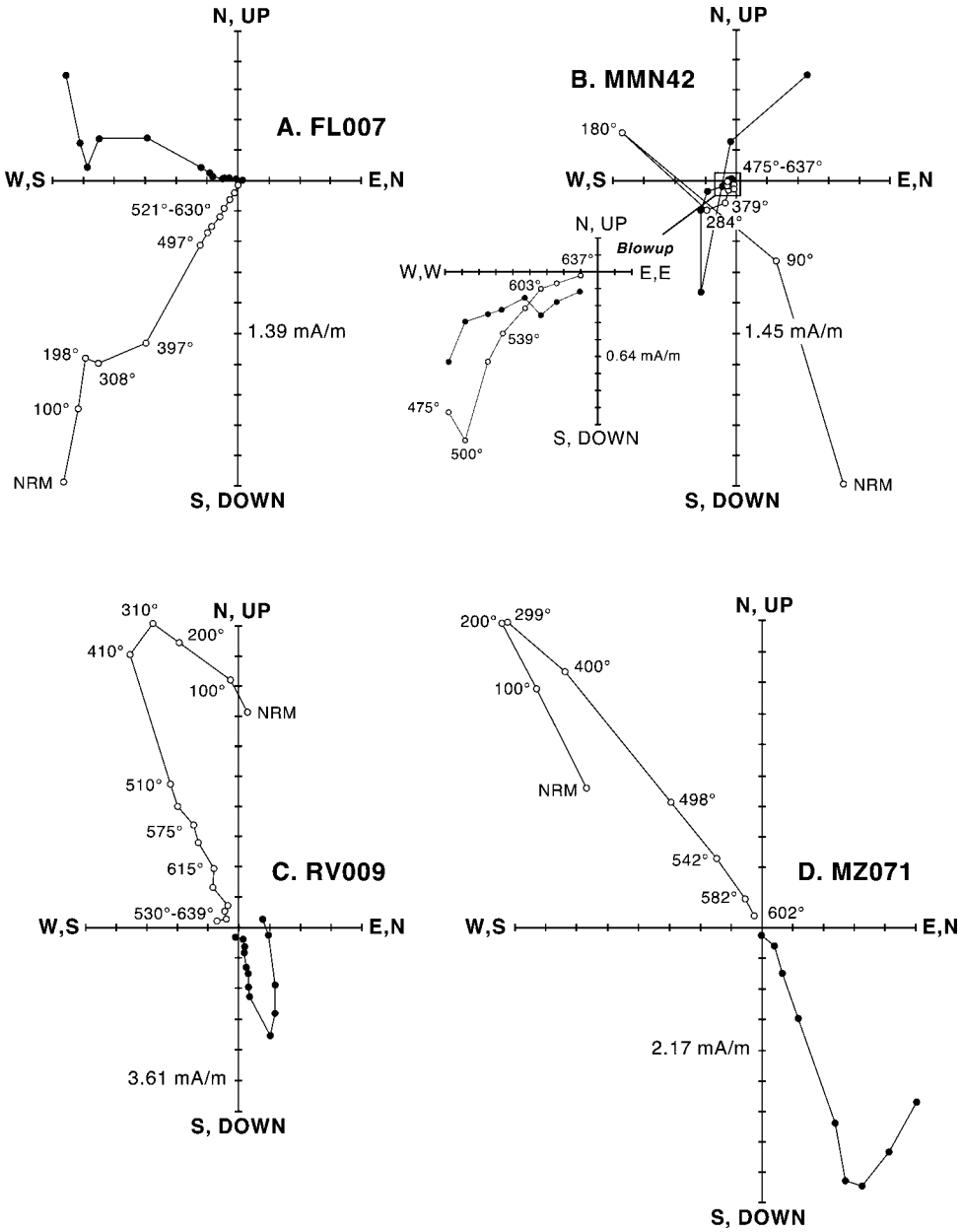


Figure 5. Orthogonal projections (in situ) of thermal demagnetization vector end points for selected chert samples: filled circles, horizontal plane; open circles, vertical plane. (A) Sample FL007 from Figline di Prato showing components A and B found at this locality. (B) Sample MMN42 from the Monte Mosca North section showing the A component and a similar B component to that at Figline (see Fig. 6A). (C) Sample RV009 from Rocchetta di Vara showing component A and a negative-inclination component B. (D) Sample MZ071 from Monte Zenone section showing components A and B similar to those at Rocchetta di Vara (see Fig. 6B).

300 °C (Fig. 5), has an in situ direction that is similar to that of the present-day field in northwestern Italy (Fig. 6A). The second component removed in most specimens (B), generally between 300 and 600 °C, is the predominant (characteristic) magnetization in the Monte Alpe Cherts specimens and has a consistent direction and polarity within each section. It has moderate positive inclinations and westerly declinations at Figline di Prato and Monte Mosca North (Fig. 6A), moderate negative inclinations and southerly declinations at Rocchetta di Vara and Monte Zenone (Fig. 6B), and steep negative inclinations at Monte Pratofiorito, Cava Termini, and Monte Mosca South (Fig. 6C). In the Tuscan Cherts section at Monte Pratofiorito and Monte Mosca South, the B component tends to be the dominant magnetization toward the base of the section in the Chert sensu strictu member and disappears toward the top of the section in the Siliceous Marlstone member (Fig. 4).

The third component (C) was isolated only in specimens from Monte Pratofiorito, Cava Termini, and Monte Mosca South, mostly between 500 and 670 °C (Table 2). Conversely with the B component, the C component is overprinted toward the base of the formation, and is the characteristic magnetization toward its top. This component has shallow inclina-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

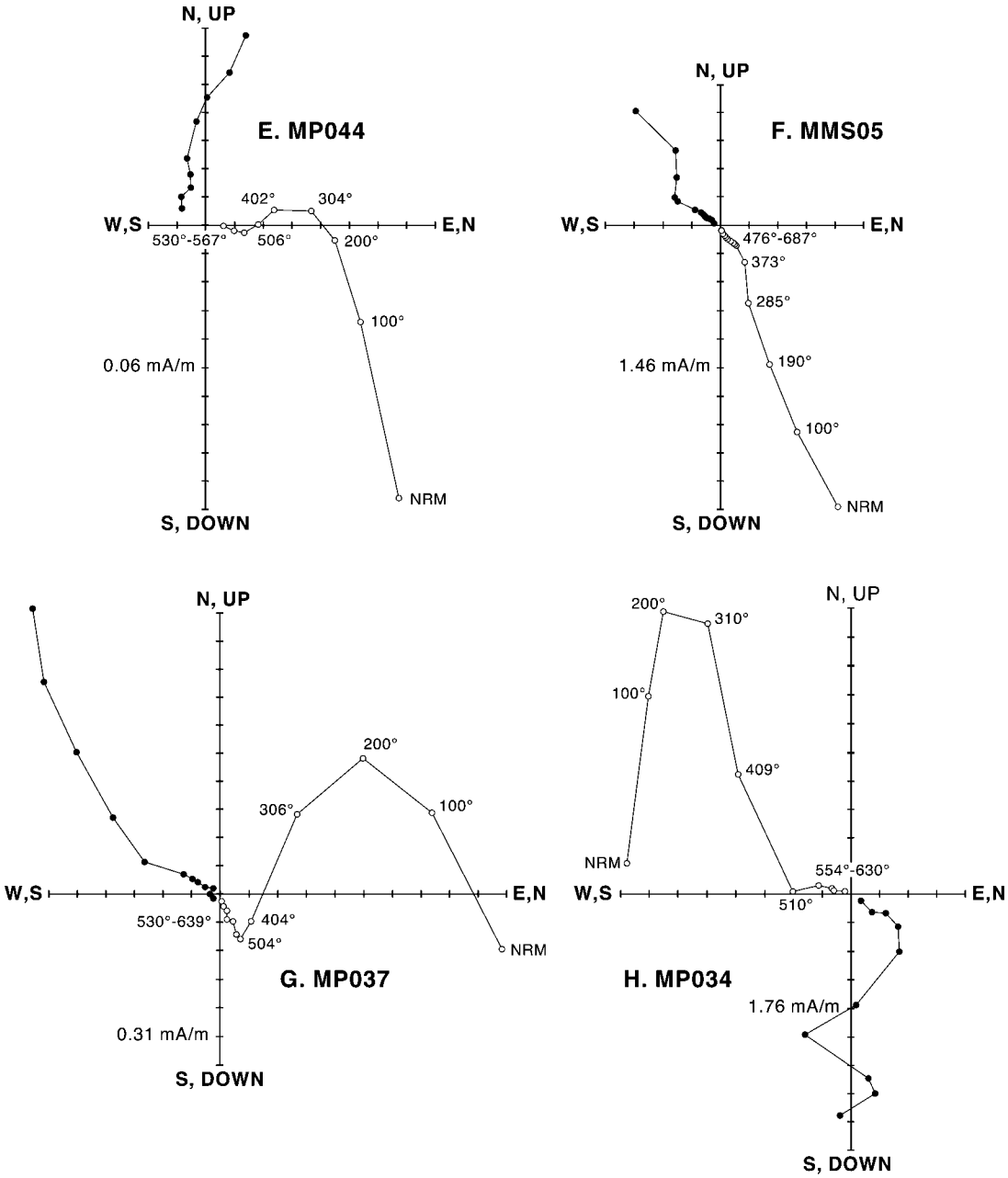


Figure 5. (*Continued*). (E) Sample MP044 from the top of Monte Pratofiorito showing component A, a much reduced component B, and a vestige of component C. (F) Sample MMS05 from nearby Monte Mosca South showing only the components A and C. (G) Sample MP037 from Monte Pratofiorito showing components A, B, and a normal C. (H) Sample MP034 from Monte Pratofiorito showing components A, B, and a reversed C.

tions (I) and declinations (D) to the west-northwest and east, indicating that both polarities of the geomagnetic field have been recorded (Fig. 6). The average direction of the westerly group is I = 26.8°, D = 287.4° (α_{95} = 6.3°, k = 24, N = 23), and the average direction of the easterly group is I = −5.8°, D = 87.4° (α_{95} = 17.0°, k = 9, N = 10). After inversion of the easterly group (I = 5.8°, D = 267.4°), these two directions are statistically different at the 95% confidence level (McFadden and Lowes, 1981) and are 28° apart (γ_c), constituting an indeterminate reversal test (McFadden and McElhinny, 1990). The failure of the reversal test is most likely due to the few and highly dispersed sample directions of the reversed-polarity group. The dispersion of this easterly group, in turn, likely results from incomplete removal of the similarly reversed B component. The presence of a polarity stratigraphy, however, supports a primary magnetization, and the mostly normal polarity is consistent with the Jurassic magnetostratigraphy determined by Channell et al. (1984) from Umbrian sections.

The Monte Pratofiorito and Cava Termini sections, however, are on opposite limbs of a regional fold (Fig. 3). The uncorrected C direction for the Monte Pratofiorito sections is I = 22.1°, D = 281.1° (α_{95} = 7.5°, k = 14, N

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

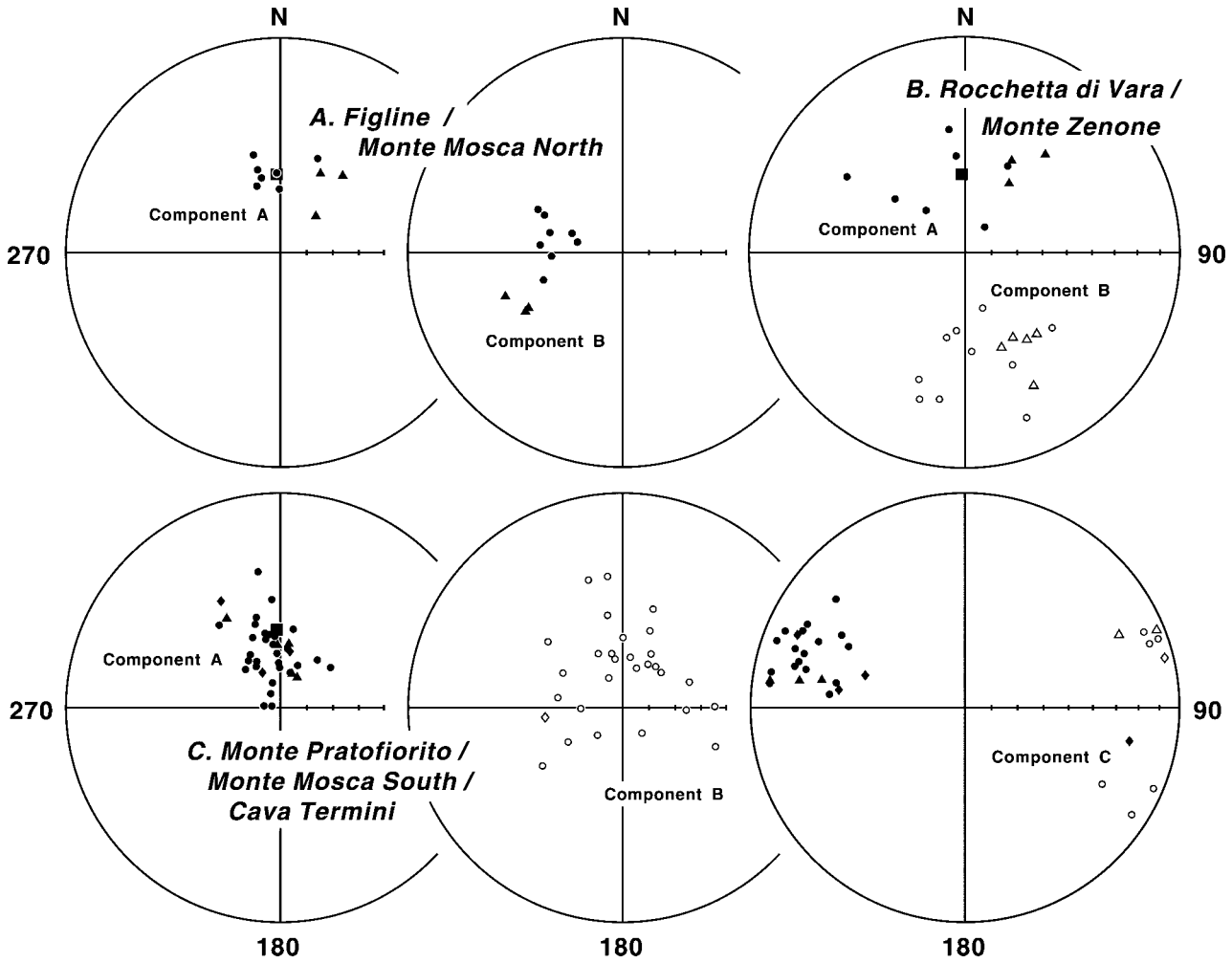


Figure 6. Equal-area plots of magnetization component directions (in situ) for chert samples: filled symbols indicate lower hemisphere points; open symbols indicate upper hemisphere points. In A, circles indicate Figline di Prato samples; triangles indicate Monte Mosca North samples. In B, circles indicate Rocchetta di Vara samples; triangles indicate Monte Zenone samples. In C, circles, triangles, and diamonds indicate Monte Pratofiorito, Monte Mosca South, and Cava Termini samples, respectively. No B components were observed in Monte Mosca South samples (see Fig. 5F). Square symbol indicates present-day field direction in northwestern Italy.

= 28), and its corrected direction is I = 21.4°, D = 281.6° (α_{95} = 7.8°, k = 13, N = 28). For the Cava Termini section, the uncorrected C direction is I = −24.1°, D = 275.0° (α_{95} = 39.3°, k = 5, N = 5), and its corrected direction is I = 18.1°, D = 278.6° (α_{95} = 32.2°, k = 7, N = 5). At the 95% confidence level, the uncorrected directions for the two limbs are different, but are the same after bedding corrections (McFadden and Lowes, 1981). Although the α_{95} for the Cava Termini samples is large, it improves upon correction and the Monte Pratofiorito and Cava Termini groups clearly come together after unfolding. This result indicates that the C magnetization predates the observed deformation, which probably occurred during late Miocene time. After corrections for the bedding tilt of each sample, the overall mean C direction is I = 21.0°, D = 281.2° (α_{95} = 7.6°, k = 12, N = 33).

DISCUSSION

Components of Magnetization

The low unblocking temperatures characterizing component A indicate that it is likely a thermoviscous overprint (viscous remnant magnetization, VRM) acquired over time under recent geomagnetic field conditions (Fig. 6). Component B from Monte Pratofiorito and Monte Mosca North is clearly a secondary magnetization and has steep negative inclinations dipping at high angles with respect to bedding (~75°; Table 1). This direction was acquired in a reversed-polarity geomagnetic field, because remagnetization most likely occurred in the Northern Hemisphere. Assuming intermediate latitudes of the Tuscan Cherts at the time of remagnetization, the chert beds must have been dipping steeply to the south in order for component B to match the expected Miocene direction of a reversed geomagnetic field. Considering the structural geology of Lima Valley (Fig. 3), a northward tilting of the chert beds after overprinting is much more likely than a vertical-axis rotation of 180°. We infer that component B was acquired by the Tuscan Cherts after the first Miocene phase of folding, when the chert beds were tilted south, and that during the second phase of folding they were tilted northward, returning the beds to their present near-horizontal position.

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 2. PALEOMAGNETIC DATA FOR THE MONTE MOSCA SOUTH, MONTE PRATOFIORITO, AND CAVA TERMINI SECTIONS

Sample	Level	I_i (°)	D_i (°)	Strike/Dip (°)/(°)	I_c (°)	D_c (°)	n	T_{Low},T_{High} (°C,°C)	λ_{VGP} (°N)	ϕ_{VGP} (°E)
Component C										
MMS06	16.6	−21.4	64.6	144/31	9.1	64.0	4	503,574	−21.7	295.7
MMS05	9.5	33.4	300.5	83/27	46.2	280.0	2	503,593	25.6	294.7
MMS04	7.3	−3.7	67.9	71/25	−4.7	69.8	1	530,608	−12.7	296.3
MMS03	6.1	32.1	280.8	84/17	35.4	269.6	2	494,715	13.2	294.8
MMS02	2.3	22.6	279.3	78/15	27.3	277.6	2	506,715	15.4	285.6
MMS01	0.8	8.6	277.9	51/25	26.1	272.2	2	545,623	11.1	288.8
MP044	70.0	22.2	310.0	262/12	13.1	312.5	1	533,550	32.4	252.6
MP043	59.0	24.4	283.5	316/08	28.5	286.9	2	509,572	22.6	279.9
MP042	52.5	17.6	298.0	284/10	14.9	300.8	1	533,550	27.2	263.4
MP041	51.0	17.2	295.4	296/12	16.9	299.1	2	512,575	26.7	265.6
MPP05	49.5	25.7	294.3	31/15	40.6	293.0	1	310,595	32.1	282.5
MPR09	47.0	30.4	280.8	220/13	18.9	283.6	1	575,630	16.5	277.9
MPR01	40.6	36.0	286.9	228/30	9.4	292.9	1	566,646	19.6	267.4
MPR05	36.4	51.4	272.5	220/20	38.4	280.9	2	103,555	22.5	289.2
MPR06	33.0	−50.0	115.2	220/23	−27.5	119.3	1	313,510	31.0	270.3
MP040	31.7	36.5	275.6	2/15	51.5	276.7	2	511,643	26.2	300.8
MPB39	30.7	21.6	288.6	336/15	32.2	294.0	2	409,632	29.2	276.7
MP039	30.0	7.3	289.6	336/15	18.0	292.0	1	534,574	22.2	271.5
MP039	30.0	−4.7	113.2	336/15	−14.7	115.1	1	614,642	23.1	267.8
MP037	27.1	38.4	297.7	146/15	29.5	289.5	1	504,569	24.8	278.5
MPB36	24.8	−9.5	70.8	208/10	−2.7	71.6	1	609,665	−12.5	294.3
MPB36	24.8	8.7	277.1	208/10	−0.7	277.4	1	569,590	5.1	274.9
MP034	20.8	−8.2	122.6	286/13	−4.3	124.0	1	554,630	25.3	256.7
MP029	17.3	17.0	289.2	313/18	23.4	295.5	1	555,660	26.7	271.2
MP027	15.0	−4.6	70.3	297/15	−15.4	72.2	1	535,555	−7.1	298.4
MP025	14.6	8.6	280.4	296/04	9.7	281.0	2	510,575	11.3	276.1
MP010	9.5	20.4	285.5	67/04	22.9	284.3	1	591,656	18.5	279.1
MP005	3.5	−10.1	67.0	251/08	10.6	68.5	1	496,601	−11.4	299.4
CT010	45.0	30.1	70.5	15/40	−4.3	75.6	1	310,659	8.8	112.0
CT009	27.5	9.5	287.0	15/40	49.5	288.0	1	510,572	32.9	292.5
CT008	26.0	0.3	279.5	15/40	40.1	277.8	4	200,596	21.1	292.2
CT004	12.5	−23.9	293.9	15/40	15.7	293.5	1	200,558	22.4	269.5
CT003	9.5	73.3	44.5	30/60	22.5	101.6	1	510,534	0.0	83.9

Notes: I_i and D_i are in situ inclination and declination of paleomagnetic directions; I_c and D_c are corrected for structural tilt; Strike/Dip, strike and dip angles of bedding, strike is 90° counterclockwise of dip direction; n is number of specimens analyzed; T_{Low},T_{High} are the low and high temperature steps of the isolated C component; λ_{VGP} and ϕ_{VGP} are latitude and longitude of virtual geomagnetic pole (VGP) corrected for structural tilt.

Component B is common in most of the Jurassic chert sections we studied in the northern Apennines (Fig. 6), specifically the Monte Alpe Cherts at Monte Zenone and Rocchetta di Vara (Table 1). Only the Tuscan Cherts at Monte Mosca North and the Monte Alpe Cherts at Figline di Prato have normal-polarity B components (Fig. 6A). The similarity of the reversed-polarity B component, although having steeper inclinations at Monte Pratofiorito, suggests that it might have been acquired simultaneously. Component B is likely a thermochemical remagnetization (see following), which is late Miocene or younger in age, and was likely acquired during regional uplift of the northern Apennine orogenic belt after thrusting of the Ligurian units over the Tuscan units.

Component C is inferred to represent a primary magnetization that was acquired by the Tuscan Cherts soon after deposition. This interpretation is supported by the presence of a polarity stratigraphy, high unblocking temperatures like those observed in other cherts (Shibuya and Sasajima, 1986; Hagstrum and Murchey, 1993), and a positive fold test. Because folding of the Tuscan units occurred during the Miocene, the fold test does not exclude the possibility that component C is a remagnetization acquired before that period. The mean inclination for the Tuscan Cherts corresponds to a paleolatitude of 11° ± 4°N (Table 2), and there is no statistically significant difference (95% confidence level; McFadden and Lowes, 1981) between the paleolatitudinal data for the Chert sensu strictu member (middle Bathonian to lower Oxfordian) and those for the Siliceous Marlstones member (middle Oxfordian to Tithonian).

Diagenesis and Acquisition of the Primary Magnetization

The paleomagnetically important iron oxides in radiolarian-rich sediments (hematite and magnetite) are derived mainly from continental sources. After deposition in a pelagic setting with other fine-grained sediments and clay, the first ferric iron to precipitate from a saturated solution is either hematite or goethite (iron hydroxide), which quickly dehydrates to hematite (Blatt, 1992). During precipitation and growth, the authigenic magnetic minerals acquire a stable remanent magnetization aligned with the local geomagnetic field (Channell et al., 1982). Color variations of cherts can be in part ascribed to different concentrations and oxidation states of these iron oxides. For example, the green color of vitreous cherts at the bottom of many radiolarian turbidite beds was probably acquired during diagenesis of almost pure radiolarian sediments. However, the green and pinkish bands within the upper siliceous marlstones of the Tuscan Cherts are more likely primary and are inferred to have been caused by reduction of iron minerals to magnetite by reducing conditions at the seafloor and/or an elevated sedimentation rate with rapid burial (see examples from the Umbrian Scaglia in Channell, 1992).

Early acquisition of the primary remanent magnetization in the Tuscan Cherts is supported by the presence of both polarities of the geomagnetic field in component C. In some specimens from the same bed, within the Chert sensu strictu member, C components having both polarities were also found (e.g., sample MP039; Table 2). Such small-scale changes in polarity probably result from a very slow sedimentation rate, which for uncompacted sediments is about 4 mm/k.y.

Petrographic analysis offers further insights into the timing and nature of diagenesis, as well as information on the recent tectonic history of the Tuscan Cherts. The beds of the Chert sensu strictu member (unit 2, Fig. 4) are composed of densely packed radiolarian tests in a dark red clayey matrix (Fig. 7A). Radiolarian tests often occur in millimeter- and submillimeter-scale laminae that developed either parallel to or at low-angles to the bedding surface; trace fossils of Thalassinoides or Planolites type are common. The lamination probably reflects sediment reworking on the seafloor by oxygenated deep-water currents. Under these conditions, organic matter was continually oxidized, favoring the precipitation of goethite and its subsequent dehydration to hematite (Channell et al., 1982). In units 3 and 4, the chert composition becomes more clayey than in unit 1.

In general, the radiolarian tests of the Chert sensu strictu member are well rounded and show only minor compactional flattening (<20%), possibly because of the earlier onset of the opal-A to opal-CT silica phase change in radiolarian-rich laminae (Kastner and Gieskes, 1983). These chert beds, which are commonly magnetically overprinted (Fig. 4), also

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

show a generation of quartz veinlets and stylolites, which probably formed during Miocene and younger deformation (Fig. 7A). Conversely, the chert beds from the Siliceous Marlstone member (Fig. 4) are generally characterized by a more prominent flattening (<50%) of poorly preserved radiolarian tests, which are often replaced by calcite, and are mixed with a prevalent groundmass of clay, microquartz, and microcalcite crystals (Fig. 7B). The flattening of the radiolarian tests is probably related to early compaction and delay in the opal-A to opal-CT transformation due to the high clay content of these beds. In thin section, the siliceous marlstone beds (only slightly overprinted; Figs. 4 and 7B) have rare stylolites and veinlets, and dissolution surfaces are uniformly distributed throughout the beds. We surmise that the two members of the Tuscan Cherts have responded differently to deformation and remagnetization because of their different rheologic properties. The chert beds of the lower member were more brittle and fractured, allowing pore fluids to circulate more easily, thus facilitating remagnetization. In contrast, the upper member behaved more plastically during deformation, thus inhibiting fluid circulation and preventing chemical remagnetization.

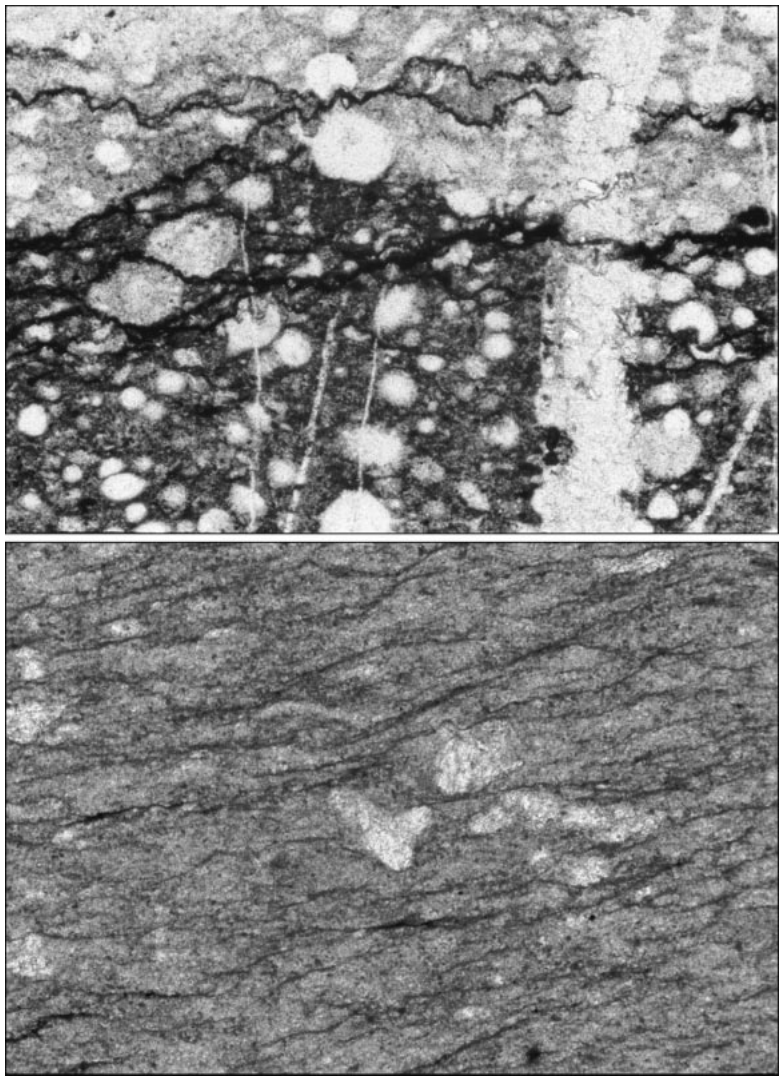


Figure 7. (A) Microphotograph of a radiolarian chert bed from unit 2 of the Tuscan Cherts at Monte Pratofiorito. The chert is a radiolarian grainstone (lower half) to packstone (upper half). Radiolarian tests show little to no flattening, and in many cases their spines are well preserved. Stylolitization and dissolution surfaces are common, as is an early generation of quartz-filled veinlets. (B) Microphotograph of a siliceous marlstone bed from unit 5 of the Tuscan Cherts at Monte Pratofiorito. Flattened radiolarian tests occur scattered within the matrix, which is composed of mixed clay, microquartz, and microcalcite. The radiolarian tests are partially replaced by microsparite. The height of both microphotographs is 350 μm.

Paleogeography and Tectonics of the Tuscan Domain

In Figure 8 the locations and trends of the Tuscan domain's main sedimentary basins, of other continental margin domains of western Tethys, and of basins floored by oceanic crust are depicted for early Late Jurassic time (ca. 155 Ma). This paleogeographical sketch has been drawn by adding paleomagnetic data available from western Tethys to those presented in this study (see caption, Fig. 8). The paleolatitude of Lima Valley (11° ± 4°N) indicates that in Middle to Late Jurassic time the Tuscan domain was at a more southerly latitude than proposed in other paleogeographic reconstructions of western Tethys (e.g., Fourcade et al., 1993).

The Tuscan Cherts were not the only radiolarian sediments of western Tethys that were deposited at low paleolatitudes. Preliminary paleomagnetic data for the Kalur Cherts of the Mirdita region in Albania indicate periequatorial paleolatitudes (Aiello and Hagstrum, 1998). The eastern Mediterranean radiolarites of Pindos-Olonos (Greece), Baër Basit (Syria), and Kaçali (Turkey) were also probably located at low latitudes (De Wever et al., 1994; Fourcade et al., 1993). In addition, low paleolatitudes were inferred for radiolarian cherts of the paleo–Pacific Ocean, such as the Mino terrane of Japan (Shibuya and Sasajima, 1986) and the Franciscan Formation of California (Hagstrum and Murchey, 1993). Thus, in Jurassic Tethys, as in the modern oceans, warm waters of tropical equatorial zones were sites for abundant growth and preservation of radiolarian plankton. The warm tropical waters of the paleo–Pacific Ocean might also have reached the western Tethys ocean, possibly by flowing through a seaway located between Africa and Adria formed by the eastward extension of a transform fault from the Ligurian-Piedmontese basin (Fig. 8; Şengör et al., 1984; Fourcade et al., 1993). According to the available paleomagnetic data, the Tuscan Cherts were more southerly than any other coeval chert sections of the Alps and Apennines (e.g., the Ruhpolding Formation of the northern Calcareous Alps, the Lombardy Chert of the southern Alps, and the Calcare con selce of Umbria), which have paleolatitudes ranging between 20° and 25°N (Fig. 8). We can speculate that, like the Tuscan domain, these latter

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 3. PALEOMAGNETIC POLES FOR ITALY, EURASIA, AND AFRICA

Section	λ_{Pole} (°N)	ϕ_{Pole} (°E)	N	R	K	A_{95} (°)	d (°)	r (°)
Tuscany								
Monte Pratofiorito								
Monte Mosca South	16	281	33	31.1041	15	8		
30° cw rotation	[37	258]						
20° cw rotation	[49	237]						
Southern Alps†								
Kimmeridgian Tithonian	39	263	11	10.7790	45	7	4 ± 8	−29 ± 8
Callovian Oxfordian*	44	270	3	2.9910	222	8	12 ± 9	−29 ± 9
Umbria§								
Late Jurassic								
Early Cretaceous	30	281	9	8.9691	259	3	9 ± 7	−12 ± 7
20° cw rotation	[44	266]						
Eurasia#								
Late Jurassic (150 Ma)	70	158	7	N.D.	N.D.	7	15 ± 8	−91 ± 8
Africa#								
Late Jurassic (150 Ma)	55	246	7	N.D.	N.D.	6	8 ± 8	−49 ± 8

Notes: λ_{Pole} and ϕ_{Pole} are latitude and longitude of paleomagnetic poles; values in brackets have been rotated into Southern Alps and African coordinates (see text); N, R, K, and A_{95} as in Table 1; d and r are the poleward displacement and counterclockwise rotation (negative values) of the sampling locality relative to the reference poles using the method of Beck (1980) as modified by Demarest (1983); N.D. indicates not determined.

*Most of the Tuscan samples are from this time interval.

†Channell et al. (1992a).

§Channell (1992).

#Besse and Courtillot (1991).

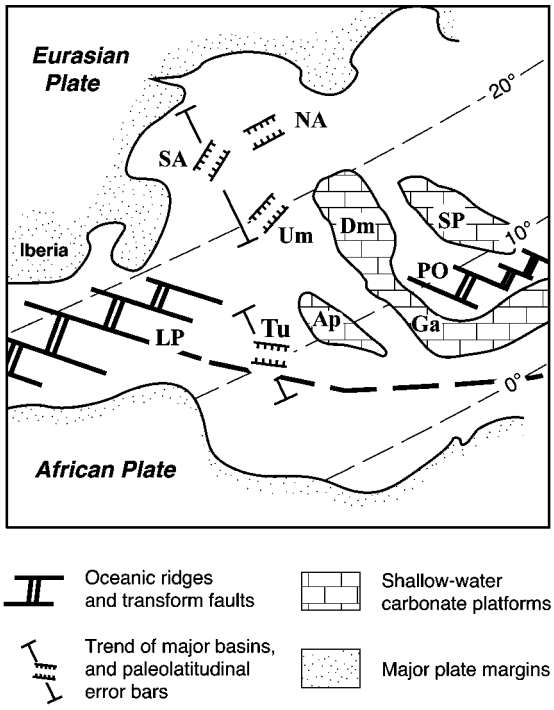


Figure 8. Paleogeographic sketch of western Tethys during early-Late Jurassic time (ca. 155 Ma). Tu—Tuscan domain; SA—southern Alps; Um—Umbrian domain; NA—northern Alps; Ap—Apulia; Ga—Gavrovo; Dm—Dalmatia; SP—Serbo-Pelagonian; LP—Ligurian-Piedmontese ocean; and PO—Pindos-Olonos ocean. Paleolatitudes of SA, NA, and Um are calculated using Channell's (1996) paleomagnetic data; paleolatitude of Africa is calculated after Besse and Courtillot (1981). The locations of Dm, Ga, SP, and PO have been modified after Fourcade et al. (1993).

basins were influenced by a warm-water current system that impinged upon the northern sectors of western Tethys by flowing through the Ligurian-Piedmontese ocean in a clockwise motion, as also suggested by DeWever et al. (1994). Modern examples of radiolarian proliferation near continental landmasses in tropical and subtropical waters have been related to the occurrence of warm water currents and coastal upwelling (e.g., Caribbean Ocean, 15°N; southern Sargasso Sea, 20°–25°N; and Somalia Basin, 10°N–10°S; Anderson, 1983).

The post-Callovian sections of the Tuscan Cherts, Monte Alpe Cherts, and other coeval sections of western Tethys and the central Atlantic show sedimentologic and lithologic evidence (e.g., occurrence of fine grained turbidites, lateral variations in bed thickness, upsection increase of clay content) for the resurgence of tectonism after a Middle Jurassic quiescent phase. By the end of Jurassic time, oceanic spreading in western Tethys had ended. To the east, closure of the Pindos-Olonos ocean caused orogenesis in the Dinarides-Hellenides, and to the north the first Alpine compressional phases had already begun. Meanwhile, the opening of the central Atlantic was progressively widening the gap between Africa and Iberia, and the paleo-Pacific and central Atlantic were connected through a seaway between the Americas (Hispanic corridor). Such a rearrangement of the major plates probably influenced the global oceanic circulation pattern, as well as that within western Tethys.

The paleomagnetic data determined for Lima Valley indicate that since Middle to Late Jurassic time the Tuscan domain underwent a counterclockwise rotation of ~90° and a northward displacement of 7°−23° with respect to Eurasia (Table 3). The counterclockwise rotation of the Tuscan domain is about 50° greater than expected if Tuscany were attached to the African plate. As suggested by Channell (1992) for the counterclockwise rotation of Umbria relative to the southern Alps (~20°), the rotation of the Tuscan domain might have occurred during Neogene deformation in the fold and thrust belt of the northern Apennines. Possibly, the Tuscan domain rotation ended before late Messinian time as indicated by paleomagnetic data from marly and clayey, postorogenic deposits of the Tuscan-Latium Tyrrhenian margin (Mattei et al., 1996; Fig. 2). In Figure 9, the poles for the Tuscan and the Umbrian domains have been rotated 30° and 20° clockwise, respectively, to restore them to southern Alps coordinates. The Tuscan pole has also been rotated clockwise another 20° to restore it to African coordinates. Although these rotations of the Tuscan pole bring it to near coincidence with the reference poles, the data also indicate significant northward displacement of the Tuscan domain with respect to the southern Alps and Africa (Table 3) using the calculation method of Beck (1980) as modified by Demarest (1983). Therefore, between Cretaceous and late Miocene time, the Tuscan domain apparently behaved as an independent tectonic terrain, which was translated a greater distance northward relative to the other continental margins of Adria. The most likely tectonic mechanism for such a latitudinal displacement is strike-slip faulting, which might have played an important, and perhaps underestimated, role in the orogenic evolution of the northern Apennines (Marroni and Treves, 1998).

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

CONCLUSIONS

Paleomagnetic study of Middle to Upper Jurassic radiolarian chert sections within the northern Apennines of Italy has provided new information for paleogeographic reconstruction of western Tethys during Middle to Late Jurassic time. Although the Monte Alpe Cherts of the Ligurian ophiolite sections of both eastern Liguria and Tuscany are probably thermochemically remagnetized, the Tuscan Cherts of the continental margin units of northern Tuscany preserve an apparently primary (C) component of remanent magnetization. The overprint components (B), observed in all sections of both the Monte Alpe Cherts and Tuscan Cherts, have reversed polarity in most of these sections. The B component is inferred to have been acquired during late Miocene regional uplift of the northern Apennine orogenic belt, which occurred after overthrusting of the Tuscan units by the Ligurian units.

In the Tuscan Cherts of Lima Valley the chemical overprint is dominant in chert beds of the lower Chert sensu strictu member and gradually disappears upsection in the Siliceous Marlstone member. Based on petrographic observations, the lower cherts were more brittle and fractured, and therefore more permeable, during deformation than was the upper, more ductile Siliceous Marlstone member. Moreover, the steep negative inclinations and northerly declinations of component B indicate that the now almost-horizontal chert beds were likely dipping steeply to the south during remagnetization, consistent with the area's tectonic history.

The C component is inferred to be a primary magnetization because it has high unblocking temperatures, a polarity stratigraphy, and a positive fold test. Its mean inclination and declination indicate a paleolatitude of 11° ± 4°N and a post-Jurassic counterclockwise rotation for the Tuscan domain of ~30° with respect to the southern Alps. These results imply a new paleogeographic scenario for western Tethys (Fig. 8), in which the Tuscan domain was more southerly than the other deep-water continental margin basins in which radiolarian-rich sediments were deposited (e.g., southern Alps and Umbria), and that transcurrent faulting might have played a significant role in the orogenic evolution of the northern Apennines.

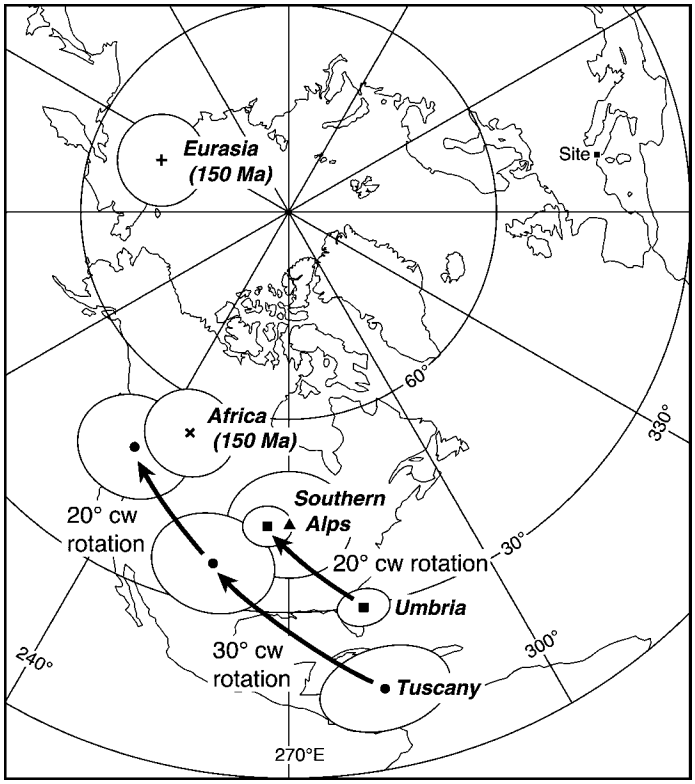


Figure 9. Late Jurassic paleomagnetic poles and their 95% confidence limits (A_{95}) for Tuscany, the southern Alps, northwest Umbria, and Africa (see Table 3). Sampling locality in northwestern Italy is also shown. Arrows indicate correction of the Tuscan and Umbrian poles for 30° and 20° clockwise (CW), respectively, and the Tuscan pole again for 20° clockwise (see text).

ACKNOWLEDGMENTS

We are grateful to Robert Garrison for critically reading the manuscript, and to Valerio Bortolotti, Gianfranco Principi, Milvio Fazzuoli, and Marco Chiari for their useful comments. Jack Hillhouse and Benita Murchey of the U.S. Geological Survey, and Edward Winterer, Don McNeill, and an anonymous reader also provided helpful and constructive reviews. This research was partially funded by Consiglio Nazionale delle Richerche—Centro di Studio di Geologia dell'Appennino e delle Catene Perimediterranee of Florence, Italy (Publication 326).

REFERENCES CITED

Abbate, E., Bortolotti, V., and Principi, G., 1980, Apennine ophiolites: A peculiar oceanic crust, *in* Rocci, G., ed., Tethyan ophiolites, 1, Western area: Ofioliti, Special Issue, v. 1, p. 59–96.

Abbate, E., Bortolotti, V., Conti, M., Marcucci, M., Principi, G., Passerini, P., and Treves, B., 1986, Apennines and Alps ophiolites and the evolution of the western Tethys: Memorie della Società Geologica Italiana, v. 31, p. 23–44.

Abbate, E., Bortolotti, V., Passerini, P., Principi, G., and Treves, B., 1994, Oceanisation processes and sedimentary evolution of the northern Apennines ophiolite suite: Memorie della Società Geologica Italiana, v. 48, p. 117–136.

Aiello, I.W., 1994, Key-beds in Tuscan Cherts of the Lima Valley and adjoining areas (Tuscan Sequence, northern Apennines): Memorie della Società Geologica Italiana, v. 48, p. 101–107.

Aiello, I.W., and Chiari, M., 1997, Biostratigraphy and facies analysis of the Middle-Late Jurassic radiolarian cherts of western Tethys: The Monte Alpe Cherts in the northern Apennines (Italy): INTERRAD VIII: Paris, Intl. Assn. Of Radiolarian Paleontologists, Abstracts, p. 3.

Aiello, I.W., and Hagstrum J.T., 1998, Paleomagnetism of Jurassic radiolarian chert sequences of the Mediterranean region: A progress report: Eos (Transactions, American Geophysical Union), v. 79, p. F224–F225.

Anderson, O.R., 1983, Radiolaria: New York, Springer, 355 p.

Barrett, T.J., 1982, Stratigraphy and sedimentology of Jurassic bedded chert overlying ophiolites in the North Apennine, Italy: Sedimentology, v. 29, p. 353–373.

Baumgartner, P.O., 1987, Age and genesis of Tethyan Jurassic radiolarites: Eclogae Geologicae Helvetiae, v. 90, p. 831–879.

Baumgartner, P.O., Bartolini, A.C., Carter, E.S., Conti, M., Cortese, G., Danelian, T., De Wever, P., Dumitrica, P., Dumitrica-Jud, R., Gorican, S., Guex, J., Hull, D.M., Kito, N., Marcucci, M., Matsuoka, A., Murchey, B., O'Dogherty, L., Savary, J., Vishnevskaya, V., Widz, D., and Yao, A., 1995, Middle Jurassic to Early Cretaceous radiolarian biochronology of Tethys based on unitary associations, *in* Baumgartner, P.O., ed., Middle Jurassic to Lower Cretaceous Radiolaria of Tethys: Occurrences, systematics, biochronology: Mémoir Géologie (Lausanne), v. 23, p. 1013–1048.

Beck, M.E., Jr., 1980, Paleomagnetic record of plate-margin processes along the western edge of North America: Journal of Geophysical Research, v. 85, p. 7115–7131.

Bernoulli, D., 1972, North Atlantic and Mediterranean Mesozoic facies: A comparison, *in* Initial reports of the Deep Sea Drilling Project, Volume 11: Washington, D.C., U.S. Government Printing Office, p. 801–871.

Bernoulli, D., and Jenkyns, H.C., 1974, Alpine, Mediterranean, and Central Atlantic Mesozoic facies in relation to the early evolution of the Tethys, *in* Dott, R.H., and Shaver, R.H., eds., Modern and ancient geosynclinal sedimentation: Society of Economic Paleontol-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

ogists and Mineralogists Special Publication 19, p. 129–160.
Besse, J., and Courtillot, V., 1991, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma: Journal of Geophysical Research, v. 96, p. 4029–4050.
Blatt, H., 1992, Sedimentary petrology: New York, W.H. Freeman and Company, 514 p.
Boccaletti, M., Elter, P., and Guazzone, G., 1971, Plate tectonics model for the development of Western Alps and the Northern Apennine: Nature, v. 234, p. 108–111.
Boccaletti, M., Coli, M., Decandia, F., Giannini, E., and Lazzarotto, A., 1980, Evoluzione dell'Appennino Settentrionale secondo un nuovo modello strutturale: Memorie della Società Geologica Italiana, v. 21, p. 359–374.
Bohrmann, G., Abelmann, A., Gersonde, R., Hubberten, H., and Kuhn, G., 1994, Pure siliceous ooze, a diagenetic environment for early chert formation: Geology, v. 22, p. 207–210.
Bosellini, A., and Winterer, E.L., 1975, Pelagic limestones and radiolarites of the Tethyan Mesozoic: A genetic model: Geology, v. 3, p. 279–282.
Channell, J.E.T., 1992, Paleomagnetic data from Umbria (Italy): Implications for the rotation of Adria and Mesozoic apparent polar wander path: Tectonophysics, v. 216, p. 365–378.
Channell, J.E.T., 1996, Paleomagnetism and paleogeography of Adria, *in* Morris, A., and Tarling, D.H., eds., Paleomagnetism and tectonics of the Mediterranean region: Geologic Society [London] Special Publication 105, p. 119–132.
Channell, J.E.T., Freeman, R., Heller, F., and Lowrie, W., 1982, Timing of diagenetic haematite growth in red pelagic limestones from Gubbio (Italy): Earth and Planetary Science Letters, v. 58, p. 189–201.
Channell, J.E.T., Lowrie, W., Pialli, P., and Venturi, F., 1984, Jurassic magnetic stratigraphy from Umbrian (Italian) land sections: Earth and Planetary Science Letters, v. 68, p. 309–325.
Channell, J.E.T., Doglioni, C., and Stoner, J.S., 1992a, Jurassic and Cretaceous paleomagnetic data from the southern Alps (Italy): Tectonics, v. 11, p. 811–822.
Channell, J.E.T., Brandner, R., Spieler, A., and Stoner, J.S., 1992b, Paleomagnetism and paleogeography of the Northern Calcareous Alps (Austria): Tectonics, v. 11, p. 792–810.
Chiari, M., Cortese, G., Marcucci, M., and Nozzoli, N., 1997, Radiolarian biostratigraphy in the sedimentary cover of ophiolites of south-western Tuscany, central Italy: Eclogae Geologicae Helvetiae, v. 90, p. 55–77.
Conti, M., and Marcucci, M., 1986, The onset of radiolarian deposition in the ophiolite sequences of the northern Apennines: Marine Micropaleontology, v. 11, p. 129–138.
Cortese, G., 1995, Radiolarian biostratigraphy of Tuscan Cherts (Tuscan Succession) from Val di Lima (Tuscany, Northern Apennine, Italy), *in* Baumgartner, P.O., ed., Middle Jurassic to Lower Cretaceous Radiolaria of Tethys: Occurrences, systematics, biochronology: Mémoir Géologie (Lausanne), v. 23, p. 813–816.
Cortesogno, L., Galbiati, B., Principi, G., and Venturelli, G., 1978, Le brecce ofiolitiche della Liguria orientale: Nuovi dati e discussione sui modelli paleogeografici: Ofioliti, v. 3, p. 261–342.
Decker, K., 1990, Plate tectonics and pelagic facies: Late Jurassic to Early Cretaceous deep-sea sediments of the Ybbsitz ophiolite unit (Eastern Alps, Austria): Sedimentary Geology, v. 67, p. 85–99.
Demarest, H.H., Jr., 1983, Error analysis for the determination of tectonic rotation from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.
Dercourt, J., Zonenshain, L.P., Ricou, L.E., Kazmin, W.G., Le Pichon, X., Knipper, A.L., Grandjacquet, C., Sbortshikov, I.M., Geyssant, J., Lepvrier, C., Pechersky, D.H., Boulin, I.J., Sibuet, J.C., Savostin, L.A., Sorokhtin, O., Westphal, M., Bazhenov, M.L., Lauer, J.P., and Biju-Duval, B., 1986, Geological evolution of the Tethys from the Atlantic to the Pamir since the Lias: Tectonophysics, v. 123, p. 241–315.
De Wever, P., Azéma, J., and Forcade, E., 1994, Radiolaires et radiolarites: Production primaire, diagénèse et paléogéographie: Bulletin des Centres de Recherches Exploration-Production, Elf Aquitaine, v. 18, p. 315–379.
Fazzuoli, M., and Sguazzoni, G., 1986, Jurassic and Cretaceous isopic zones in the Tuscan domain: Memorie della Società Geologica Italiana, v. 31, p. 59–84.
Fazzuoli, M., Sani, F., Ferrini, G., Garzonio, C.A., Sguazzoni, G., Becarelli, S., Burchietti, G., and Mannori, G., 1998, Geologia del Nucleo Mesozoico della Val di Lima (Provincie di Pistoia e Lucca, Appennino Settentrionale): Note Illustrative della Carta Geologica (scala 1:25 000): Bollettino della Società Geologica Italiana, v. 117, p. 479–535.
Fisher, R.A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, ser. A, v. 217, p. 295–305.
Folk, R.L., and McBride, E.F., 1978, Radiolarites and their relations to subjacent 'oceanic crust' in Liguria, Italy: Journal of Sedimentary Petrology, v. 48, p. 1069–1102.
Fourcade, E., Azéma, J., Cecca, F., Dercourt, J., Bellion, Y., Ricou, E., Sandulescu, M., and Vrielynck, B., 1993, Atlas de la Tethys Pelaeoenvironmental Maps, *in* Dercourt, J., et al., eds., BEICIP-FRANLAB: Rueil-Mailmaison, France, maps.
Garrison, R.E., 1974, Radiolarian cherts, pelagic limestones and igneous rocks in eugeosynclinal assemblages, *in* Hsü, K.J., and Jenkyns, H.C., eds., Pelagic sediments on land and under the Sea: International Association of Sedimentologists Special Publication 1, p. 367–399.
Garrison, R.E., and Fischer, A.G., 1969, Deep-water limestones and radiolarites of the Alpine Jurassic, *in* Friedman, G.M., ed., Depositional environments in carbonate rocks: Society of Economic Paleontologists and Mineralogists Special Publication 14, p. 20–56.
Gawlick, H.J., Frisch, W., Vecsei, A., Steiger, T., and Böhm, F., 1999, The change from rifting to thrusting in the Northern Calcareous Alps as recorded in Jurassic sediments: Geologische Rundschau, v. 87, p. 644–657.
Goll, R.M., and Bjørklund, K.R., 1971, Radiolaria in surface sediments of the North Atlantic Ocean: Micropaleontology, v. 25, p. 365–396.
Hagstrum, J.T., 1990, Remagnetization and northward coastwise transport of Franciscan Complex rocks, northern California: A reinterpretation of the paleomagnetic data: Tectonics, v. 9, p. 1221–1233.
Hagstrum, J.T., 1993, North American Jurassic APW: The current dilemma: Eos (Transactions, American Geophysical Union), v. 74, p. 65–69.
Hagstrum, J.T., and Murchey, B.L., 1993, Deposition of Franciscan Complex cherts along the paleoequator and accretion to the American margin at tropical paleolatitudes: Geological Society of America Bulletin, v. 105, p. 766–778.
Hagstrum, J.T., and Murchey, B.L., 1996, Paleomagnetism of Jurassic radiolarian chert above the Coast Range ophiolite at Stanley Mountain, California, and implications for its paleogeographic origins: Geological Society of America Bulletin, v. 108, p. 643–652.
Hagstrum, J.T., and Sedlock, R.L., 1992, Paleomagnetism of Mesozoic red chert from Cedros Island and the San Benito Islands, Baja California, Mexico, revisited: Geophysical Research Letters, v. 19, p. 329–332.
Hagstrum, J.T., Murchey, B.L., and Bogar, R.S., 1996, Equatorial origin for Lower Jurassic radiolarian chert in the Franciscan Complex, San Rafael Mountains, southern California: Journal of Geophysical Research, v. 101, p. 613–626.
Heath, G.R., 1974, Dissolved silica in deep-sea sediments, *in* Hay, W.W., ed., Studies in paleo-oceanography: Society of Economic Paleontologists and Mineralogists Special Publication 20, p. 77–93.
Jenkyns, H.C., and Winterer, E.L., 1982, Paleoceanography of Mesozoic ribbon radiolarites: Earth and Planetary Science Letters, v. 60, p. 352–375.
Kälin, O., Patacca, E., and Renz, O., 1979, Jurassic pelagic deposits from southeastern Tuscany: Aspects of sedimentation and new biostratigraphic data: Eclogae Geologicae Helvetiae, v. 72, p. 715–762.
Kastner, M., and Gieskes, J.M., 1983, Opal-A to opal-CT transformation: A kinetic study, *in* Iijima, A., et al., eds., Siliceous deposits in the Pacific region: Developments in Sedimentology 36: Amsterdam, Elsevier, p. 211–227.
Kastner, M., Keene, J.B., and Gieskes, J.M., 1977, Diagenesis of siliceous oozes−I. Chemical controls on the rate of opal-A to opal-CT transformation—An experimental study: Geochimica et Cosmochimica Acta, v. 41, p. 1041–1059.
Kirschvink, J.L., 1980, The least-squares line and plane and analysis of paleomagnetic data: Royal Astronomical Society Geophysical Journal, v. 62, p. 699–718.
Lemoine, M., Tricart, P., and Boillot, G., 1987, Ultramafic and gabbroic ocean floor of the Ligurian Tethys (Alps, Corsica, Apennine): In search of a genetic model: Geology, v. 15, p. 622–625.
Marroni, M., and Treves, B., 1998, Hidden terranes in the northern Apennines, Italy: A record of Late Cretaceous−Oligocene transpressional tectonics: Journal of Geology, v. 106, p. 149–162.
Mattei, M., Kissel, C., Sagnotti, L., Funiciello, R., and Faccenna, C., 1996, Lack of late Miocene to present rotation in the Northern Tyrrhenian margin (Italy): A constraint on geodynamic evolution, *in* Morris, A., and Tarling, D.H., eds., Paleomagnetism and tectonics of the Mediterranean region: Geological Society [London] Special Publication 105, p. 141–147.
McBride, E.F., and Folk, R.L., 1979, Features and origin of Italian Jurassic radiolarites deposited on continental crust: Journal of Sedimentary Petrology, v. 49, p. 837–868.
McFadden, P.L., and Lowes, F.J., 1981, The discrimination of mean directions drawn from Fisher distributions: Royal Astronomical Society Geophysical Journal, v. 67, p. 19–33.
McFadden, P.L., and McElhinny, M.W., 1990, Classification of the reversal test in paleomagnetism: Geophysical Journal International, v. 103, p. 725–729.
Molina, J.M., O'Dogherty, L., Sandoval, J., and Vera, J.A., 1999, Jurassic radiolarites in a Tethyan continental margin (Subbetic, southern Spain): Paleobathymetric and biostratigraphic considerations: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 150, p. 309–330.
Nishbet, E.G., and Price, I., 1974, Siliceous turbidites: Bedded cherts as redeposited ocean ridge-derived sediments, *in* Hsü, K.J., and Jenkyns, H.C., eds., Pelagic sediments on land and under the sea: International Association of Sedimentologists Special Publication 1, p. 351–366.
Pisciotto, K.A., 1981, Diagenetic trends in the siliceous facies of the Monterey Shale in the Santa Maria region, California: Sedimentology, v. 28, p. 547–571.
Principi, G., and Treves, B., 1984, Il sistema corso-appenninico come prisma di accrezione, Riflessi sul problema generale del limite Alpi-Appennini: Memorie della Società Geologica Italiana, v. 28, p. 549–576.
Robertson, A.H.F., Dixon, J.E., Brown, S., Collins, A., Morris, A., Pikett, E.A., Sharp, I., and Ustaömer, T., 1996, Alternative tectonic models for the late Paleozoic−early Tertiary development of Tethys in the eastern Mediterranean region, *in* Morris, A., and Tarling, D.H., eds., Paleomagnetism and tectonics of the Mediterranean region: Geological Society [London] Special Publication 105, p. 239–264.
Şengör, A.M.C., Yilmaz, Y., and Süngürlu, O., 1984, Tectonics of the Mediterranean Cimmerides: Nature and evolution of the western termination of Palaeo-Tethys, *in* Dixon, J.E., and Robertson, A.H.F., eds., The geological evolution of the eastern Mediterranean: Geological Society [London] Special Publication 17, p. 77–112.
Shibuya, H., and Sasajima, S., 1986, Paleomagnetism of red cherts: A case study in the Inuyama area, central Japan: Journal of Geophysical Research, v. 91, p. 14105–14116.
Van der Voo, R., 1993, Paleomagnetism of the Atlantic, Tethys, and Iapetus Oceans: Cambridge, Cambridge University Press, 411 p.
Van Fossen, M.C., and Kent, D.V., 1990, High-latitude paleomagnetic poles from Middle Jurassic plutons and moat volcanics in New England and the controversy regarding Jurassic apparent polar wander for North America: Journal of Geophysical Research, v. 95, p. 503–516.
Winterer, E.L., 1998, Paleobathymetry of Mediterranean Tethyan Jurassic pelagic sediments: Memorie della Società Geologica Italiana, v. 53, p. 97–131.

MANUSCRIPT RECEIVED BY THE SOCIETY AUGUST 17, 1999
REVISED MANUSCRIPT RECEIVED APRIL 6, 2000
MANUSCRIPT ACCEPTED MAY 16, 2000

Printed in the USA